Simplifying your distributed cloud infrastructure with MicroCloud

WHITEPAPER
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Executive summary

A steep learning curve, operational complexity, deployment hurdles, and unpredictable costs – these are familiar challenges for the growing number of enterprises adopting distributed hybrid and multi-cloud strategies. This whitepaper offers a deep dive into these challenges, and showcases how organizations can simplify the management of distributed cloud infrastructures with Canonical's MicroCloud.

This paper details the necessity for a streamlined approach to cloud environments, focusing on MicroCloud's minimalist architecture and automation features. Translating theory into action, it also provides a step-by-step guide for designing and deploying your own MicroCloud. By adopting MicroCloud, businesses can reduce operational burdens and costs, enabling them to concentrate on driving innovation and efficiently managing workloads across diverse environments. The recommendations emphasize the importance of adopting modular, open architectures and automation to achieve an agile and competitive cloud strategy.

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Introduction

Virtualization transformed the way businesses operate data centers, laying the foundation for the cloud computing revolution. By abstracting physical hardware into virtual machines, companies could increase efficiency, flexibility, and scalability. As a result, cloud computing evolved as a natural next step, providing on-demand computing resources through public and private clouds.

Today, most organizations rely on a hybrid cloud model, with some workloads running on-premise and others on public cloud platforms. In fact, according to a recent study by Flexera [1], 89% of enterprises have embraced multi-cloud strategies, balancing the benefits of both public and private cloud services. Yet, in the past few years, infrastructure has evolved further into a distributed model. Companies no longer rely solely on centralized cloud infrastructures but instead distribute workloads across multiple locations, clouds, and even edge environments.

According to Gartner [2], by 2025, 75% of enterprise data will be processed outside of a traditional data center or cloud, in various distributed environments. This shift requires organizations to adopt simpler, more optimized, and easier-to-operate infrastructure to support a multi-cloud, distributed computing paradigm.

The goal of this whitepaper is to provide readers with a comprehensive understanding of the complexities associated with distributed cloud infrastructure, and present MicroCloud as a viable solution. Readers can expect to gain insights into the specific challenges organizations face in cloud adoption and optimization, as well as practical recommendations for addressing those issues. By exploring the functionalities and advantages of MicroCloud, this whitepaper aims to equip IT decision-makers and cloud architects with the knowledge needed to simplify their cloud operations and enhance their infrastructure strategies.

1 *Flexera 2024 State of the Cloud Report, https://info.flexera.com/CM-REPORT-State-of-the-Cloud*

2 *Gartner, What Edge Computing Means for Infrastructure and Operations Leaders https://www.gartner.com/smarterwithgartner/what-edge-computing-means-for-infrastructure-and-operations-leaders*

SECTION

1

Key challenges in distributed cloud infrastructure

As enterprises increasingly shift towards hybrid and multi-cloud, they encounter several significant challenges that can impede their adoption and optimization of distributed cloud infrastructure. This section delves into four core obstacles: the steep learning curve, operational complexity, deployment hurdles, and unpredictable costs.

The need for skilled personnel to navigate complex cloud environments, especially in hybrid and multi-cloud setups, is a primary concern, with companies struggling to manage security, compliance, and system integration. Additionally, scaling private clouds, managing geographically distributed data, and controlling cloud spend add layers of complexity that often result in inefficiencies and higher costs. Understanding these challenges is crucial for businesses aiming to leverage the full potential of cloud technologies while maintaining flexibility, security, and cost control.

Learning curve

The shift to cloud computing requires a substantial investment in training and upskilling. According to IBM[3], 69% of respondents mentioned a lack of skills as a major challenge in cloud adoption. This shortage of skilled personnel has become one of the primary obstacles for businesses moving toward hybrid and multi-cloud environments.

Cloud service providers offer various training programs and certifications, but the learning curve remains steep. The skills required span multiple domains, including cloud security, governance, automation, and container orchestration (e.g., Kubernetes). While multi-cloud adoption offers resilience and flexibility, it amplifies the complexity of skills required. Each cloud platform, public or private, has its own set of tools, APIs, and configurations, forcing teams to juggle multiple learning tracks. Moreover, the rapid pace of innovation in cloud technologies means that ongoing education and adaptability are essential for success. With cloud-native architectures growing in complexity, businesses face the challenge of maintaining an agile workforce capable of handling both legacy infrastructure and new cloud environments.

Operations complexity

Cloud operations become especially complicated in multi-cloud and hybrid setups due to the need for interoperability between disparate systems. This complexity extends across several areas.

Managing diverse ecosystems (on-premises, private, and public clouds) requires the integration of systems that often do not natively communicate well with each other. Differences in cloud services and APIs increase operational burdens. Compliance and Security is another important operational concern. With sensitive workloads spread across multiple environments, maintaining security and regulatory compliance is a daunting task. Compliance with GDPR, HIPAA, and other regulations is exponentially more difficult in distributed environments. This is one of the main reasons companies cannot host all of their data in public cloud environments and need a multi-cloud approach.

System maintenance is another closely related area. To ensure security and compliance, systems need to be maintained and regularly patched and upgraded to avoid any security incidents. However, regular maintenance can be a challenge in complex environments and can take a lot of effort and time, as well as lead to outages if not handled properly.

Finally, data management is something that needs to be considered. With infrastructure becoming more distributed, data management also becomes complex. Ensuring data consistency across different environments while maintaining low latency and high availability presents technical challenges, particularly in geographically distributed cloud setups. Handling edge computing workloads only exacerbates these challenges.

3 *IBM Transformation Index: State of Cloud,* *https://www.ibm.com/thought-leadership/institute-business-value/en-us/report/transformation-index*

Deployment challenges

Private cloud deployments often entail significant upfront investments in hardware, networking equipment, and storage. Companies typically turn to specialized consulting firms for assistance in configuring these environments, which drives up costs and timelines.

Furthermore, scaling private cloud environments adds another layer of difficulty. Traditional on-premise setups are generally slower to scale than public cloud environments, which provide on-demand resources. This becomes an operational bottleneck when businesses experience rapid growth or seasonal demand spikes.

Certain industries, like healthcare and finance, face specific challenges related to compliance. Data sovereignty and industry-specific regulations can further delay private cloud deployments, as custom configurations must be set up to meet these requirements.

Unpredictable costs

Cost unpredictability in public cloud environments remains one of the top concerns for enterprises. Cloud providers typically use a pay-as-you-go pricing model, which offers flexibility but can quickly spiral out of control without proper monitoring. According to Flexera's 2024 State of the Cloud Report[4], 84% of enterprises rank managing public cloud spend as a significant issue.

One of the main contributors to cost unpredictability is "cloud sprawl," where businesses deploy services across various cloud platforms without fully optimizing resource usage. Cloud sprawl often results in underutilized or idle resources, driving up costs unnecessarily. Respondents to Respondents of Flexera's 2024 State of the Cloud Report reported their public cloud waste is 27%. This can be an issue on the private cloud side as well. Due to deployment complexities outlined above, companies may wish to deploy larger infrastructure to cover for future growth, leading to wasted resources and expensive licencing.

Vendor lock-in is another cost-related concern. Many organizations find themselves tied to a single cloud provider's proprietary services, making it difficult to switch vendors or adopt a multi-cloud strategy. This lack of flexibility not only inflates costs but also limits innovation. Enterprises using a single cloud provider often express concerns over vendor lock-in, fearing the inability to negotiate pricing or pivot to alternative solutions.

4 *Flexera 2024 State of the Cloud Report, https://info.flexera.com/CM-REPORT-State-of-the-Cloud*

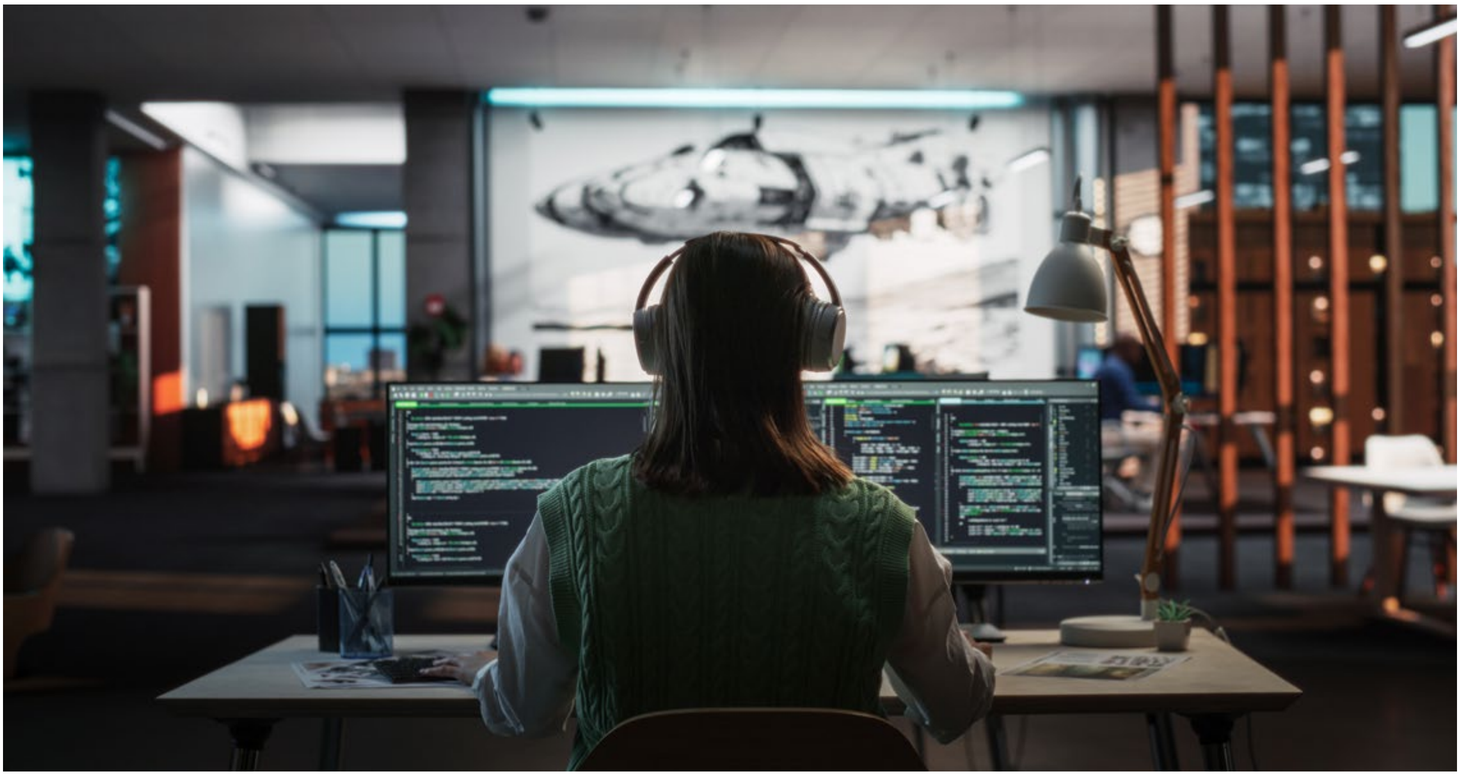
SECTION

2

Reducing the complexity of cloud infrastructure

The central theme that connects all the challenges outlined above is complexity – so how can organizations bring simplicity to their distributed cloud infrastructure? Best practices revolve around three core principles: optimized design, streamlined operations, and simplified deployment.

By adopting modular and open cloud architectures, companies can enhance flexibility and scalability while avoiding vendor lock-in. Simplified deployment frameworks, driven by automation and pre-configured templates, allow for faster and more cost-effective cloud implementations. Additionally, integrating automation and standardized tools in day-to-day operations reduces operational overhead and improves resource efficiency. These approaches not only cut costs and time but also position organizations to be more agile and innovative in an increasingly competitive market.



Optimized design

Decades of technologial evolution have helped businesses understand what works best in cloud infrastructure. A well-optimized design focuses on simplicity, flexibility, and scalability, enabling businesses to avoid vendor lock-in and complex proprietary solutions. Enterprises are now favoring modular and open architectures, which allow for greater flexibility when migrating workloads between environments.

With the rise of IoT, edge computing has become increasingly critical in modern infrastructure designs. Optimized cloud architectures incorporate edge components to handle data processing closer to the source of data generation. This reduces latency and ensures real-time processing capabilities. According to IDC, the global edge computing market is expected to reach $378 Billion in 2028[5], making it a vital consideration in distributed infrastructure design.

Simplified deployment

Simplified deployment plays a pivotal role in optimizing both time and cost efficiency when establishing private cloud infrastructure. By streamlining the deployment process, organizations can achieve a more agile and cost-effective implementation. Simplified deployment frameworks often encompass automated provisioning, pre-configured templates, and standardized processes that significantly reduce the complexities associated with traditional infrastructure setups.

These streamlined methodologies minimize manual intervention, thereby decreasing the likelihood of errors and accelerating the deployment timeline. The use of automation tools and pre-configured solutions enables rapid scaling and efficient resource management, which not only shortens the time to operational readiness but also curtails associated labor costs.

Furthermore, simplified deployment reduces the need for extensive training and specialized expertise, allowing IT teams to allocate resources more strategically and focus on higher-value tasks. This operational efficiency translates to substantial cost savings and allows organizations to realize the benefits of their private cloud infrastructure more swiftly. Ultimately, adopting a streamlined deployment approach enhances overall productivity and positions organizations to better leverage their cloud resources in a competitive landscape.

5 *IDC, The Worldwide Edge Spending Guide, https://www.idc.com/getdoc.jsp?containerId=IDC_P39947*

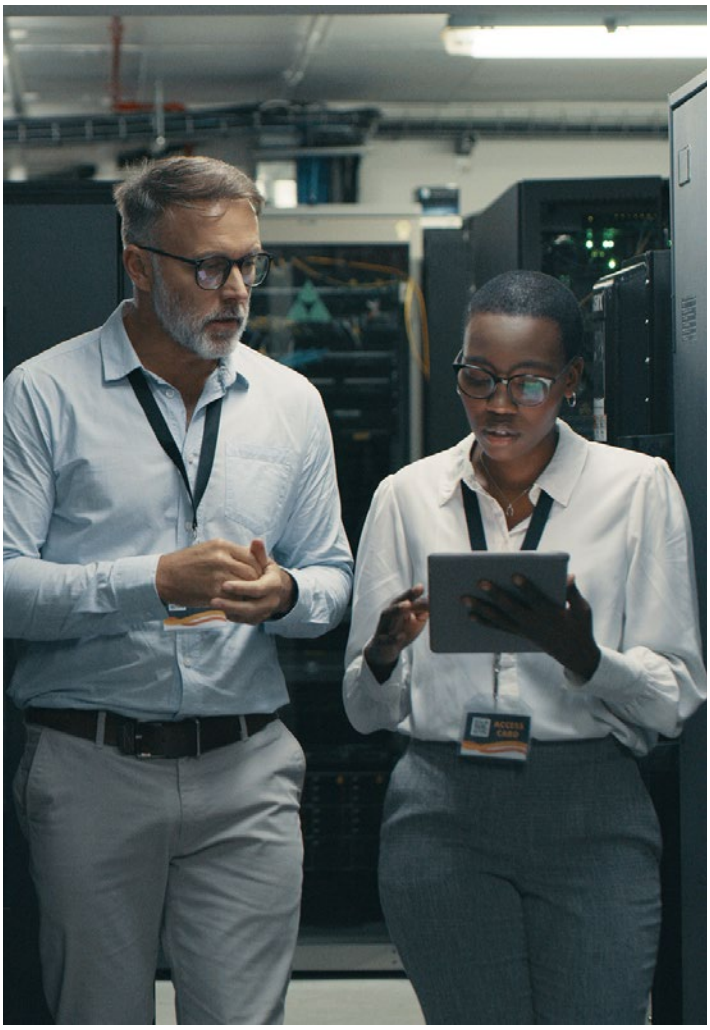
Streamlined operations

Simplified cloud infrastructure operations offer significant business value by enhancing operational efficiency and driving strategic advantages. By streamlining cloud management processes, organizations can achieve more effective resource utilization, reduced operational overhead, and accelerated time-to-market for new initiatives.

A simplified approach to cloud infrastructure operations often involves leveraging automation, standardized configurations, and integrated management tools. These practices reduce the complexity and manual effort required for day-to-day management, resulting in a more responsive and agile IT environment. This operational efficiency translates into lower costs associated with maintenance, troubleshooting, and human resource requirements, freeing up capital for investment in growth and innovation.

Moreover, streamlined cloud operations contribute to improved reliability and performance. By minimizing the potential for human error and optimizing resource allocation, organizations can enhance system uptime and service quality, which is crucial for maintaining competitive advantage and customer satisfaction.

The business value extends beyond operational efficiency; simplified cloud infrastructure enables organizations to scale more effectively in response to changing market demands. This scalability supports the rapid deployment of new services and applications, fostering agility and innovation.



SECTION 3

MicroCloud: addressing cloud infrastructure challenges

Canonical's MicroCloud offers a powerful solution to the challenges outlined above. MicroClouds are small-scale clouds optimized for repeatable and reliable remote deployments.

MicroCloud is designed to deliver the benefits of private cloud infrastructure while addressing the complexity, cost, and operational hurdles faced by companies managing hybrid and distributed environments. Here's how MicroCloud addresses these issues:

Optimized design

MicroCloud offers a minimalistic yet comprehensive infrastructure design tailored for edge and distributed environments. With a focus on simplicity, MicroCloud allows businesses to discard unnecessary elements found in traditional data centers, optimizing their infrastructure. By delivering a lightweight, easy-to-deploy architecture, MicroCloud enables businesses to run cloud workloads with greater efficiency.

MicroCloud has three main components that, when combined, provide comprehensive feature coverage with minimal complexity. LXD, a lightweight open source virtualization platform, is at the heart of MicroCloud. Rather than having different modules that need to be enabled, LXD features around deploying, configuring and operating workloads are built-in by default for an out-of-the-box experience. When extended with software-defined storage and networking, LXD turns into a MicroCloud. All components are packaged using snaps, meaning that all dependencies are containerized together with the application, significantly reducing external dependencies and simplifying the architecture.

Simplified operations

MicroCloud is built with automation at its core. Through its built-in management capabilities, including simplified upgrades, patching, and monitoring, MicroCloud reduces the operational complexity associated with traditional cloud environments. With snaps, the packaging mechanism used by all MicroCloud components, software updates are automatic and resilient.

Operations can be done via the web UI, CLI, or the REST API, providing options for engineers with various skill sets. Projects, profiles, and cluster groups allow for simplified configuration in bulk, saving time and minimizing complexities.

In addition, Canonical's tools such as Landscape and Livepatch further enhance this by automating routine tasks, minimizing downtime, and improving system performance. With MicroCloud, businesses can shift their focus from managing infrastructure to driving innovation through workloads and applications.

Cost efficiency

Canonical's MicroCloud is designed to offer cost predictability and optimization, addressing both private and public cloud challenges. By reducing upfront investment requirements for private cloud deployments, and providing transparent, subscription-based support models, MicroCloud allows organizations to better manage their IT budgets. Its lightweight infrastructure reduces hardware demands, further contributing to lower overall costs.

MicroCloud is a fully open source solution, involving no prohibitive licensing fees. Optional enterprise support—available under the Ubuntu Pro subscription— covers not only MicroCloud, but also thousands of open source packages. Its transparent pricing and extensive coverage provide peace of mind for companies while ensuring the highest levels of security and compliance, for a fraction of the cost in comparison to proprietary tools.

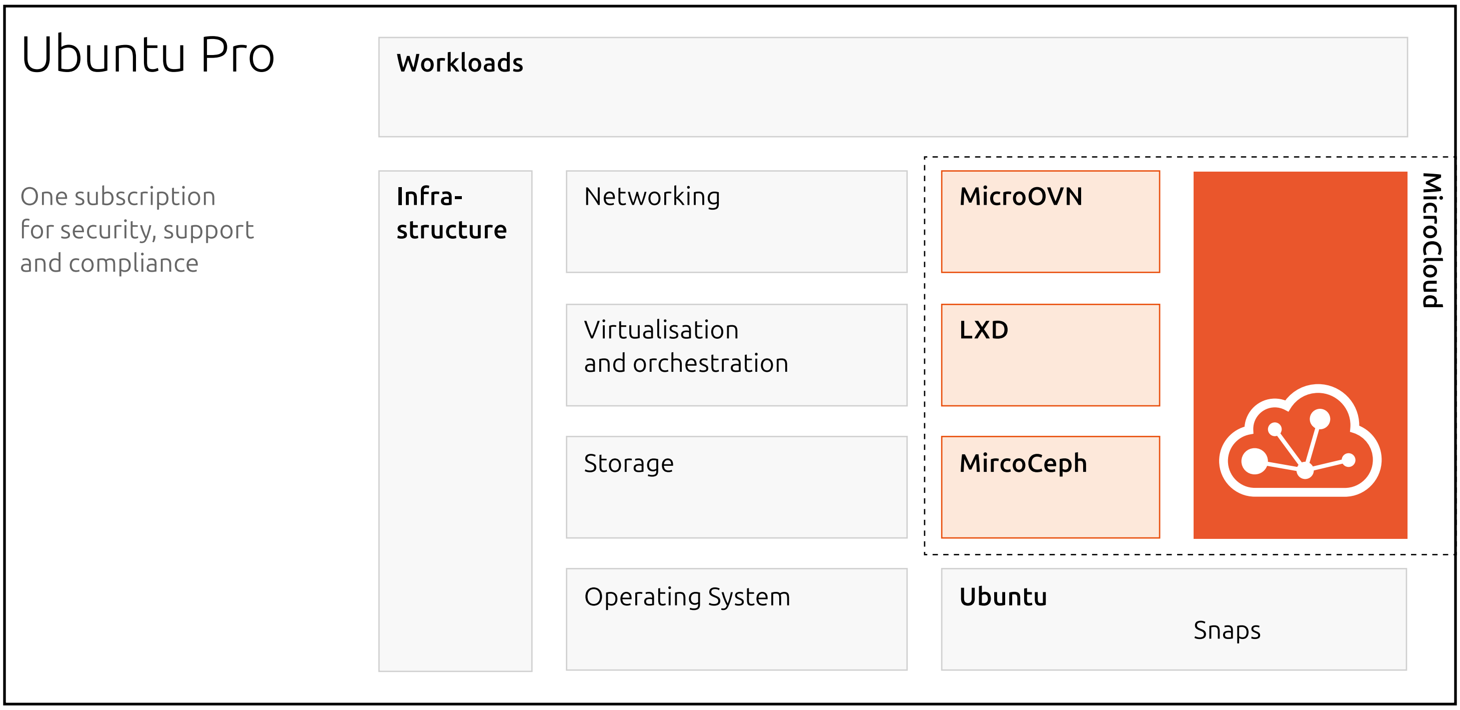
MicroCloud scales easily and thus enables organizations to enjoy the financial benefits of both private and public clouds without the complexity or unpredictable costs.

Simplified deployment

MicroCloud dramatically simplifies deployment through its intuitive, automated setup process. With out-of-the-box compatibility for bare-metal and virtualized environments, MicroCloud enables organizations to deploy fully operational clouds within minutes, not weeks. It eliminates the need for expensive consulting services and significantly reduces deployment lead times, empowering businesses to rapidly scale their operations without major capital investment.

Additionally, MicroCloud's flexible architecture supports a range of use cases, from on-premises cloud environments to edge computing, enabling organizations to scale their operations seamlessly. You can start with a single-node deployment for testing, and then easily scale to a highly available, secure, and redundant cloud.

"Deploy fully operational clouds in minutes" is a bold claim, but you can put it to the test yourself by deploying your own MicroCloud.



SECTION 4

Deploy your own MicroCloud

What you need

MicroCloud minimum hardware requirements are fairly low, as it is lightweight and its control plane doesn't utilize many resources. However, you should consider whether this is a test or production environment, as well as what kind of workloads you will run on top, and make sure your hardware can support that. Users can choose whether they want to use local or distributed storage or both of them, as well as choose whether they want to use distributed networking or just use a simple bridge network. This gives flexibility for users to easily set up their cloud environment in the way that best suits their use case. Below, we outline some high-level recommendations for test/dev and production environments.

Test/Dev environment

MicroCloud supports single-member deployments which are great for testing and development, as well as for non-critical deployments where data redundancy isn't mandatory.

We typically recommend at least 8GiB of RAM per machine.

For single-member deployments, especially for test/dev environments, local storage is often sufficient, and it is fast and convenient. If this is sufficient for your use case, you can skip setting the distributed storage.

If you do choose to set up distributed storage, you can do so on a single member, but it won't have the recommended replication configuration which ensures high availability. For HA, and the ability to recover a cluster should something go wrong, you need a minimum of three members, and at least three disks on at least three different machines for Ceph.

For networking, MicroCloud requires two dedicated network interfaces: one for intra-cluster communication and one for external connectivity. To allow for external connectivity, MicroCloud requires an uplink network that supports broadcast and multicast.

For single-member deployments that are not expected to grow, OVN might not be necessary, and this step can then be skipped.

Test/Dev environment *continued*

Additional members can be added anytime by running `microcloud add` command on the new member. This will restart the initialization dialogue and can be used to add services that previously weren't configured. For example, an initial single member MicroCloud could utilize just local storage, the user can then add two new members and when running `microcloud add` on them they can then choose to deploy distributed storage which will also set it on the initial member.

Production environments

As production environments typically run many business-critical applications, we would recommend additional resources to ensure a robust and redundant environment.

In terms of resources, we recommend at least 32GB of RAM per machine.

For a 3-member cluster, we recommend 9 NVMe disks, 3/physical host. One would be used for the OS, 1 for local storage, and 1 for distributed storage. Disks for local and remote storage should be free of any partition/file system information.

For networking, two dedicated network interfaces are still required, so dual port network cards with a minimum of 10Gb are recommended for good throughput, more if the applications you are running require low latency.

While three cluster members are sufficient for high availability, losing one member can leave the customers exposed to a split-brain situation when running just on two members. That's why having an extra member is recommended when running critical applications.

For better performance, users can choose to designate a separate network for Ceph internal traffic, as well as an underlay network for distributed networking. In addition, users can choose to encrypt their Ceph storage during the deployment.

Installation process

To prepare your servers for the MicroCloud initialization process, all servers you intend to use should be provisioned with an Ubuntu LTS. For details on how to do that, follow the tutorial outlined in the Ubuntu documentation.

Once your servers are ready, install the required snaps on all servers by running: `sudo snap install microcloud lxd microceph microovn`

With all snaps installed, you can proceed to the initialization step.

Initialization process

Complete the following steps to initialize MicroCloud:

1. SSH to one of the members, then enter the following command: `sudo microcloud init`

   You will then get a list of questions, and your answers will be used for configuring your cluster.

2. Select whether you want to set up more than one member. If you want to deploy a single member MicroCloud, it will skip the `trust-establishment-session` as no more members need to be included.

   Additional members can always be added at a later point in time.

To allow several members of MicroCloud joining the final cluster, in both the interactive and anon-interactive method, each member is running one half of the trust establishment session to trust the other side.

Each trust establishment session has one initiator and one to many joiners. In the case of the interactive mode, the side that runs the `microcloud init` command becomes the initiator. The other side becomes the joiner by running `microcloud join`.

If required, MicroCloud uses `mDNS (multicast DNS)` to automatically detect a so-called initiator on the network. This method works in physical networks, but it is usually not supported in a cloud environment. Instead, you can specify the address of the initiator to not require using mDNS.

The scan is limited to the local subnet of the network interface you select when choosing an address for MicroCloud's internal traffic.

3. Select the IP address that you want to use for MicroCloud's internal traffic. MicroCloud automatically detects the available addresses (IPv4 and IPv6) on the existing network interfaces and displays them in a table. You must select exactly one address.

4. If deploying a multiple member MicroCloud, on all the other machines, enter the following command and repeat the address selection: `sudo microcloud join` In each terminal select an address for MicroCloud's internal traffic. When prompted enter the passphrase in each terminal and return to the first member.

5. Select the members that you want to add to the MicroCloud cluster. MicroCloud displays all members that have reached out during the trust establishment session. Make sure that all members that you select have the required snaps installed.

6. Select whether you want to set up local storage. If you choose yes, configure the local storage:

    6.1. Select the disks that you want to use for local storage. You must select exactly one disk from each machine.

    6.2. Select whether you want to wipe any of the disks. Wiping a disk will destroy all data on it.

7. Select whether you want to set up distributed storage (using MicroCeph). If you choose `yes`, configure the distributed storage:

    7.1. Select the disks that you want to use for distributed storage. You must select at least three disks.

    7.2. Select whether you want to wipe any of the disks. Wiping a disk will destroy all data on it.

    7.3. Select whether you want to encrypt any of the disks. Encrypting a disk will store the encryption keys in the Ceph key ring inside the Ceph configuration folder.

    7.4. You can choose to optionally set up a CephFS distributed file system.

8. Select either an IPv4 or IPv6 CIDR subnet for the Ceph internal traffic. You can leave it empty to use the default value, which is the MicroCloud internal network.

9. Select whether you want to set up distributed networking (using MicroOVN). If you choose `yes`, configure the distributed networking:

    9.1. Select the network interfaces that you want to use. You must select one network interface per machine.

    9.2. If you want to use IPv4, specify the IPv4 gateway on the uplink network (in CIDR notation) and the first and last IPv4 address in the range that you want to use with LXD.

    9.3. If you want to use IPv6, specify the IPv6 gateway on the uplink network (in CIDR notation).

    9.4. If you chose to set up distributed networking, you can choose to setup an underlay network for the distributed networking: If you choose yes, configure the underlay network:

        9.4.1 Select the network interfaces that you want to use.

        9.4.2 You must select one network interface with an IP address per machine.

10. MicroCloud now starts to bootstrap the cluster. Monitor the output to see whether all steps are completed successfully.

Once the initialisation process is complete, your MicroCloud should be fully operational and you can start using it.

Here is an example output of a `microcloud init` CLI dialogue for the initiating side:

```
root@micro1:~#microcloud init
Do you want to set up more than one cluster member? (yes/no) [default=yes]: yes
Select an address for MicroCloud's internal traffic:
Space to select; enter to confirm; type to filter results.
Up/down to move; right to select all; left to select none.
       +----------------------+--------+
       |       ADDRESS        | IFACE  |
       +----------------------+--------+
> [X]  | 203.0.113.169        | enp5s0 |
  [ ]  | 2001:db8:d:100::169  | enp5s0 |
       +----------------------+--------+
 Using address "203.0.113.169" for MicroCloud

Use the following command on systems that you want to join the cluster:

 microcloud join

When requested enter the passphrase:

 koala absorbing update dorsal

Verify the fingerprint "5d0808de679d" is displayed on joining systems.
Waiting to detect systems ...
Select the systems that should join the cluster:
Space to select; enter to confirm; type to filter results.
Up/down to move; right to select all; left to select none.
       +---------+---------------+--------------+
       |  NAME   |    ADDRESS    | FINGERPRINT  |
       +---------+---------------+--------------+
> [x]  | micro3  | 203.0.113.171 | 4e80954d6a64 |
  [x]  | micro2  | 203.0.113.170 | 84e0b50e13b3 |
  [x]  | micro4  | 203.0.113.172 | 98667a808a99 |
       +---------+---------------+--------------+

 Selected "micro1" at "203.0.113.169"
 Selected "micro3" at "203.0.113.171"
 Selected "micro2" at "203.0.113.170"
 Selected "micro4" at "203.0.113.172"

Would you like to set up local storage? (yes/no) [default=yes]: yes
Select exactly one disk from each cluster member:
Space to select; enter to confirm; type to filter results.
Up/down to move; right to select all; left to select none.
       +----------+---------------+----------+------+------------------------------------------------------+
       | LOCATION |     MODEL     | CAPACITY | TYPE |                         PATH                         |
       +----------+---------------+----------+------+------------------------------------------------------+
  [x]  | micro1   | QEMU HARDDISK | 10.00GiB | scsi | /dev/disk/by-id/scsi-0QEMU_QEMU_HARDDISK_lxd_local1  |
  [ ]  | micro1   | QEMU HARDDISK | 20.00GiB | scsi | /dev/disk/by-id/scsi-0QEMU_QEMU_HARDDISK_lxd_remote1 |
  [x]  | micro2   | QEMU HARDDISK | 10.00GiB | scsi | /dev/disk/by-id/scsi-0QEMU_QEMU_HARDDISK_lxd_local2  |
  [ ]  | micro2   | QEMU HARDDISK | 20.00GiB | scsi | /dev/disk/by-id/scsi-0QEMU_QEMU_HARDDISK_lxd_remote2 |
  [x]  | micro3   | QEMU HARDDISK | 10.00GiB | scsi | /dev/disk/by-id/scsi-0QEMU_QEMU_HARDDISK_lxd_local3  |
  [ ]  | micro3   | QEMU HARDDISK | 20.00GiB | scsi | /dev/disk/by-id/scsi-0QEMU_QEMU_HARDDISK_lxd_remote3 |
> [x]  | micro4   | QEMU HARDDISK | 10.00GiB | scsi | /dev/disk/by-id/scsi-0QEMU_QEMU_HARDDISK_lxd_local4  |
       +----------+---------------+----------+------+------------------------------------------------------+

Select which disks to wipe:
Space to select; enter to confirm; type to filter results.
```

Here is an example output of a `microcloud init` CLI dialogue for the initiating side (continued):

```
Up/down to move; right to select all; left to select none.
       +----------+--------------+----------+------+------------------------------------------------------+
       | LOCATION |     MODEL    | CAPACITY | TYPE |                         PATH                         |
       +----------+--------------+----------+------+------------------------------------------------------+
> [ ]  | micro1   | QEMU HARDDISK | 10.00GiB | scsi | /dev/disk/by-id/scsi-0QEMU_QEMU_HARDDISK_lxd_local1  |
  [ ]  | micro2   | QEMU HARDDISK | 10.00GiB | scsi | /dev/disk/by-id/scsi-0QEMU_QEMU_HARDDISK_lxd_local2  |
  [ ]  | micro3   | QEMU HARDDISK | 10.00GiB | scsi | /dev/disk/by-id/scsi-0QEMU_QEMU_HARDDISK_lxd_local3  |
  [ ]  | micro4   | QEMU HARDDISK | 10.00GiB | scsi | /dev/disk/by-id/scsi-0QEMU_QEMU_HARDDISK_lxd_local4  |
       +----------+--------------+----------+------+------------------------------------------------------+

 Using "/dev/disk/by-id/scsi-0QEMU_QEMU_HARDDISK_lxd_local3" on "micro3" for local storage pool
 Using "/dev/disk/by-id/scsi-0QEMU_QEMU_HARDDISK_lxd_local4" on "micro4" for local storage pool
 Using "/dev/disk/by-id/scsi-0QEMU_QEMU_HARDDISK_lxd_local1" on "micro1" for local storage pool
 Using "/dev/disk/by-id/scsi-0QEMU_QEMU_HARDDISK_lxd_local2" on "micro2" for local storage pool

Would you like to set up distributed storage? (yes/no) [default=yes]: yes
Unable to find disks on some systems. Continue anyway? (yes/no) [default=yes]: yes
Select from the available unpartitioned disks:
Space to select; enter to confirm; type to filter results.
Up/down to move; right to select all; left to select none.
       +----------+--------------+----------+------+------------------------------------------------------+
       | LOCATION |     MODEL    | CAPACITY | TYPE |                         PATH                         |
       +----------+--------------+----------+------+------------------------------------------------------+
> [x]  | micro1   | QEMU HARDDISK | 20.00GiB | scsi | /dev/disk/by-id/scsi-0QEMU_QEMU_HARDDISK_lxd_remote1 |
  [x]  | micro2   | QEMU HARDDISK | 20.00GiB | scsi | /dev/disk/by-id/scsi-0QEMU_QEMU_HARDDISK_lxd_remote2 |
  [x]  | micro3   | QEMU HARDDISK | 20.00GiB | scsi | /dev/disk/by-id/scsi-0QEMU_QEMU_HARDDISK_lxd_remote3 |
       +----------+--------------+----------+------+------------------------------------------------------+

Select which disks to wipe:
Space to select; enter to confirm; type to filter results.
Up/down to move; right to select all; left to select none.
       +----------+--------------+----------+------+------------------------------------------------------+
       | LOCATION |     MODEL    | CAPACITY | TYPE |                         PATH                         |
       +----------+--------------+----------+------+------------------------------------------------------+
> [ ]  | micro1   | QEMU HARDDISK | 20.00GiB | scsi | /dev/disk/by-id/scsi-0QEMU_QEMU_HARDDISK_lxd_remote1 |
  [ ]  | micro2   | QEMU HARDDISK | 20.00GiB | scsi | /dev/disk/by-id/scsi-0QEMU_QEMU_HARDDISK_lxd_remote2 |
  [ ]  | micro3   | QEMU HARDDISK | 20.00GiB | scsi | /dev/disk/by-id/scsi-0QEMU_QEMU_HARDDISK_lxd_remote3 |
       +----------+--------------+----------+------+------------------------------------------------------+

 Using 1 disk(s) on "micro1" for remote storage pool
 Using 1 disk(s) on "micro2" for remote storage pool
 Using 1 disk(s) on "micro3" for remote storage pool

Do you want to encrypt the selected disks? (yes/no) [default=no]: no
Would you like to set up CephFS remote storage? (yes/no) [default=yes]:  yes
What subnet (either IPv4 or IPv6 CIDR notation) would you like your Ceph internal traffic on? [default:
203.0.113.0/24]:
What subnet (either IPv4 or IPv6 CIDR notation) would you like your Ceph public traffic on? [default:
203.0.113.0/24]:
Configure distributed networking? (yes/no) [default=yes]:  yes
Select an available interface per system to provide external connectivity for distributed network(s):
Space to select; enter to confirm; type to filter results.
Up/down to move; right to select all; left to select none.
       +----------+--------+----------+
       | LOCATION | IFACE  |   TYPE   |
       +----------+--------+----------+
> [x]  | micro2   | enp6s0 | physical |
  [x]  | micro3   | enp6s0 | physical |
```

Here is an example output of a `microcloud init` CLI dialogue for the initiating side (continued):

```
    [x]  | micro1   | enp6s0 | physical |
    [x]  | micro4   | enp6s0 | physical |
         +----------+--------+----------+

 Using "enp6s0" on "micro3" for OVN uplink
 Using "enp6s0" on "micro1" for OVN uplink
 Using "enp6s0" on "micro2" for OVN uplink
 Using "enp6s0" on "micro4" for OVN uplink

Specify the IPv4 gateway (CIDR) on the uplink network (empty to skip IPv4): 192.0.2.1/24
Specify the first IPv4 address in the range to use on the uplink network: 192.0.2.100
Specify the last IPv4 address in the range to use on the uplink network: 192.0.2.254
Specify the IPv6 gateway (CIDR) on the uplink network (empty to skip IPv6): 2001:db8:d:200::1/64
Specify the DNS addresses (comma-separated IPv4 / IPv6 addresses) for the distributed network (default:
192.0.2.1,2001:db8:d:200::1):
Configure dedicated underlay networking? (yes/no) [default=no]:

Initializing new services
 Local MicroCloud is ready
 Local LXD is ready
 Local MicroOVN is ready
 Local MicroCeph is ready
Awaiting cluster formation ...
 Peer "micro2" has joined the cluster
 Peer "micro3" has joined the cluster
 Peer "micro4" has joined the cluster
Configuring cluster-wide devices ...
MicroCloud is ready
```

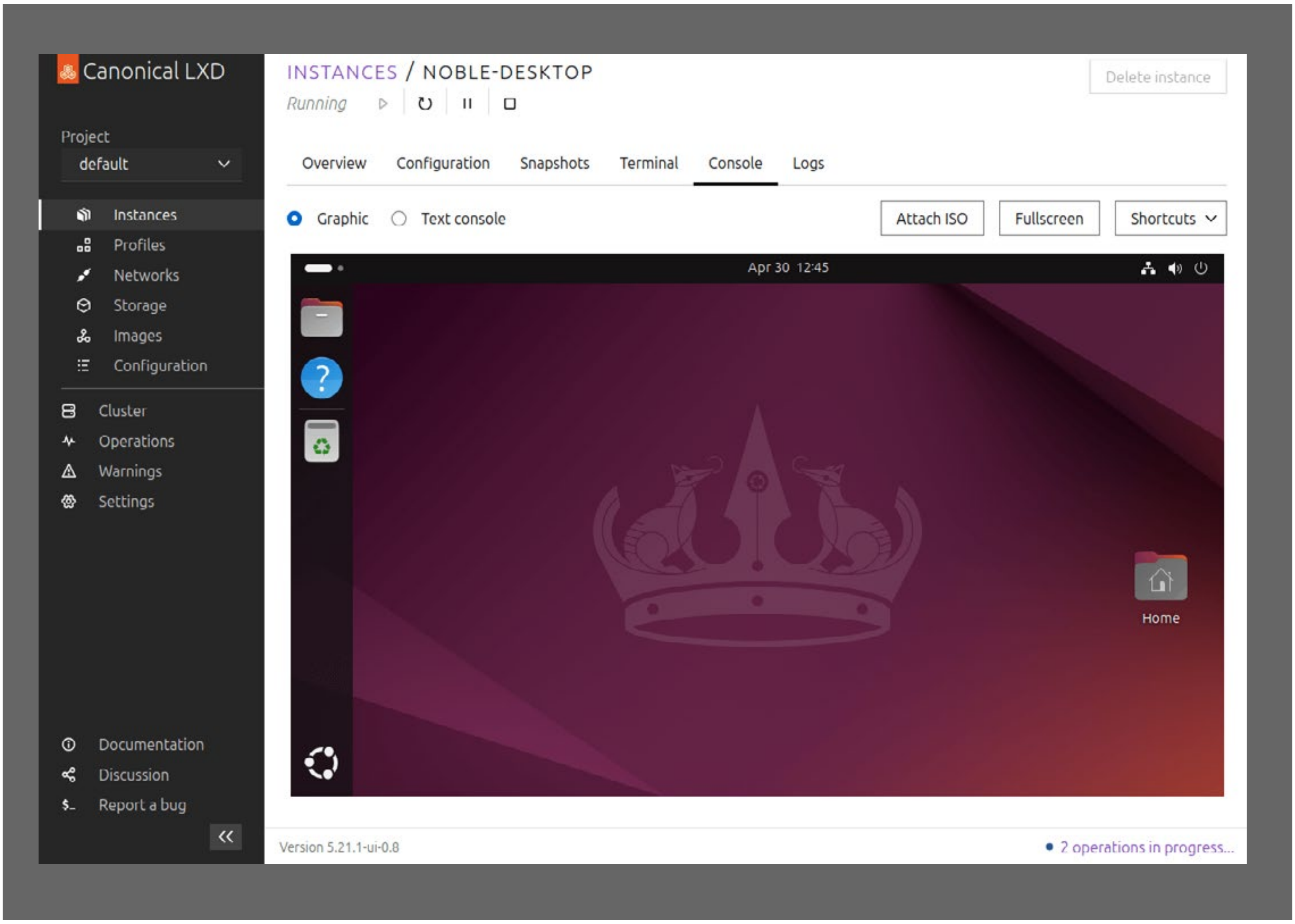
Here is an example output for one of the joining sides `(micro02):`

```
root@micro1:~#microcloud init
Select an address for MicroCloud's internal traffic:

 Using address "203.0.113.170" for MicroCloud

Verify the fingerprint "84e0b50e13b3" is displayed on the other system.
Specify the passphrase for joining the system: koala absorbing update dorsal
Searching for an eligible system ...

 Found system "micro01" at "203.0.113.169" using fingerprint "5d0808de679d"

Select "micro02" on "micro01" to let it join the cluster

 Received confirmation from system "micro01"

Do not exit out to keep the session alive.
Complete the remaining configuration on "micro01" ...
Successfully joined the cluster
```

Accessing the UI

LXD UI is used to interact with the entire MicroCloud. To enable it follow the how-to outlined in the documentation.

You are now ready to work with your MicroCloud.



Conclusion

As cloud computing evolves toward more distributed and hybrid models, the complexity and cost of managing these environments present significant challenges. Traditional cloud infrastructure, while powerful, can be burdensome due to its steep learning curves, complex operations, and unpredictable expenses.

MicroCloud provides a compelling solution by simplifying cloud infrastructure with an optimized design, streamlined deployment, and automated operations. By reducing both the operational and financial barriers to cloud adoption, MicroCloud enables businesses to focus on their workloads, rather than the complexities of infrastructure management. Whether operating in public, private, or hybrid cloud environments, MicroCloud offers a path forward to reducing the complexity of distributed computing.

Ultimately, Canonical's MicroCloud presents a cost-effective, simplified infrastructure solution that meets the demands of the modern, distributed cloud landscape, empowering organizations to innovate faster while keeping costs in check.

Have a use case in mind?

Get in touch ›

ADDITIONAL RESOURCES

1. https://canonical.com/microcloud
2. https://canonical.com/lxd
3. https://canonical.com/solutions/infrastructure
4. https://ubuntu.com/engage/university-ontario-francais-microcloud-case-study
5. https://ubuntu.com/engage/guide-to-micro-clouds
6. https://ubuntu.com/engage/vmware-migration-scenarios

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